

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - II

Roll No. : S025	Name: Abhishek Pal
Class : S.T.I.T	Batch : 1
Date of Assignment : 22/07/26	Date/Time of Submission : 22/07/26

Questions:

2a. Write a python code to create nested tuples.

```
student = ("Abhishek", 25, ("BSC IT", "Semester III"), 100)

print("Name :", student[0])
print("Roll no:", student[1])
print("Department :", student[2][0])
print("Semester :", student[2][1])
print("Percentage :", student[-1])
print("Abhishek S025")
```

```
Name : Abhishek
Roll no: 25
Department : BSC IT
Semester : Semester III
Percentage : 100
Abhishek S025
```

2b. Write a python code to sort the nested tuple using sorted () function.

```
data = ("Sahil", "Babu", "Aku")
sorted_data = sorted(data, key = lambda x:x[2])

print(sorted_data)
print("Abhishek S025")
```

```
['Babu', 'Sahil', 'Aku']
Abhishek S025
```

2c. Write a python code to copy or clone list.

```
list1 = ("Abhishek", "Babu", "Aku")
list2 = list1

print(list2)
print("Abhishek S025")
```

```
('Abhishek', 'Babu', 'Aku')
Abhishek S025
```

2d. Write a python code to check immutability property of python tuples.

```
tup=( (1,2) , (2,1) , (3,2) )

tup[1]=(1,8)

print(tup)

print("Abhishek S025")
```

```
Traceback (most recent call last):
  File "C:/Users/itlab/Desktop/S025 - Abhishek/PDS Prac 2.py", line 3, in <module>
    tup[1]=(1,8)
TypeError: 'tuple' object does not support item assignment
```

Curiosity Questions:

Q1. Define a tuple. How is it different from a list? Give one example.

Q2. Write a Python program to create the following tuples:

- a) An empty tuple
- b) A tuple with integers only
- c) A tuple with mixed data types
- d) A nested tuple (tuple inside another tuple)

Q3. Write a program to access elements of a tuple using indexing and slicing.

Q4. Create a tuple and find the following:

- Length of the tuple
- Maximum and minimum value (if all are numbers)
- Count of a specific item

Q5. Write a program to convert a list into a tuple.

Q6. Write a program to sort a tuple of numbers in ascending order.

Q7. Write a program to show that tuples are immutable (cannot be changed).

Q8. Write a program to clone (copy) a list and print both original and copied lists.

Q9. Write the output of the following code:

```
t = (1, 2, 3, 4, 5)
```

```
print(t[1:4])
```

Q10. Create a tuple of student marks: (78, 85, 92, 88, 76). Write a program to:

- Find average marks
- Display marks above 80